

Requirements Analysis and Specification

A **requirement** is a description of what a system should do. It defines:

- *What to build* (functional requirements).
- *How the system should behave* (non-functional requirements).

Requirements act as a **bridge between customer needs and system design**.

Types of Requirements

1. Functional Requirements

- Define *what* the system should do.
- Example: “The system shall allow users to log in using a username and password.”

2. Non-functional Requirements

- Define *quality attributes* of the system.
- Example: Performance, security, reliability, usability, scalability.

3. Domain Requirements

- Derived from the application domain.
- Example: In a banking system, interest calculation rules.

Requirements Analysis Process

1. Feasibility Study

- Is the project economically, technically, and operationally feasible?

2. Requirement Elicitation (Gathering)

- Interviews, surveys, observation, brainstorming, prototyping.

3. Requirement Analysis

- Resolving conflicts, checking completeness, consistency, feasibility.

4. Requirement Specification

- Documenting requirements in a structured form (SRS document).

5. Requirement Validation

- Ensuring requirements are correct and reflect user needs.
- Techniques: Reviews, prototyping, test-case generation.

Software Requirements Specification (SRS)

An **SRS document** is the official contract between client and developer.

Contents of SRS (IEEE 830 standard):

1. Introduction

- Purpose, scope, definitions.

2. Overall description

- Product perspective, functions, constraints.

3. Functional requirements

4. Non-functional requirements

5. External interface requirements (UI, hardware, software).

6. Appendices & glossary

Qualities of a Good SRS:

- Correct, Complete, Unambiguous, Verifiable, Consistent, Modifiable, Traceable.

Software Project Planning:

Software Project Planning = Process of **defining scope, estimating effort, cost, resources, and scheduling activities** before starting development.

Goal: Deliver software on time, within budget, and with required quality.

Objectives of Project Planning

- Define project scope clearly.
- Identify resources (people, tools, hardware, software).
- Estimate cost, time, and effort.
- Identify risks and mitigation strategies.
- Create a project schedule.

Activities in Project Planning

1. Scope Definition

- Clearly define project boundaries, goals, and deliverables.

2. Effort Estimation

- Use models to estimate development effort.
- Common methods:
 - LOC (Lines of Code) estimation
 - Function Point Analysis (FPA)
 - COCOMO Model (Constructive Cost Model)

3. Resource Planning

- People: Skills required, team size.
- Tools: Development & testing environments.
- Hardware/Software resources.

4. Scheduling

- Break project into tasks.
- Tools:
 - **Gantt Chart** (time vs tasks)
 - **PERT/CPM Chart** (dependency analysis, critical path).

5. Risk Management

- Identify risks (technical, financial, schedule, requirement changes).
- Plan mitigation strategies.

6. Quality Planning

- Define quality objectives.
- Plan reviews, testing, and standards compliance.

Project Estimation Techniques

1. Lines of Code (LOC) Based Estimation

- Effort is estimated based on expected lines of source code.
- Limitation: Language-dependent, inaccurate for early stages.

2. Function Point Analysis (FPA)

- Measures software size by counting **functions delivered** (inputs, outputs, files, interfaces).
- Language-independent, better for early stages.

3. COCOMO Model

- Developed by Barry Boehm.
- Formula:

$$\text{Effort} = a \times (\text{KLOC})^b$$

where a and b depend on project type (organic, semi-detached, embedded).

- Estimates: Effort (person-months), Development time, Staffing.

Project Scheduling Tools

1. Gantt Chart

- Bar chart showing timeline of tasks.
- Simple visualization.

2. PERT/CPM (Program Evaluation and Review Technique / Critical Path Method)

- Graph-based representation.
- Identifies task dependencies and critical path (longest path in project).

Importance of Project Planning

- Prevents cost overruns and delays.
- Helps manage risks effectively.
- Provides clear roadmap for execution.
- Increases customer confidence.